

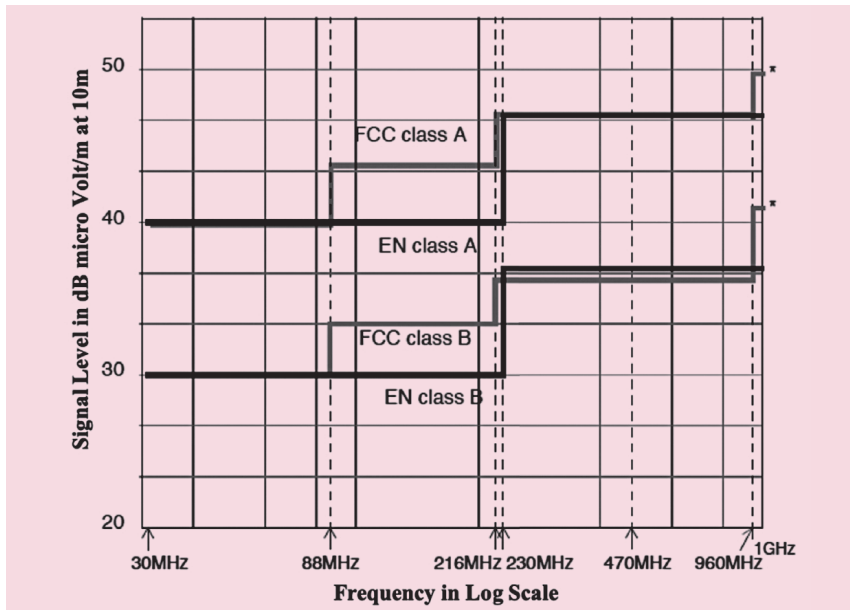


Fig. 2: Radiated and conducted emission levels of FCC standards

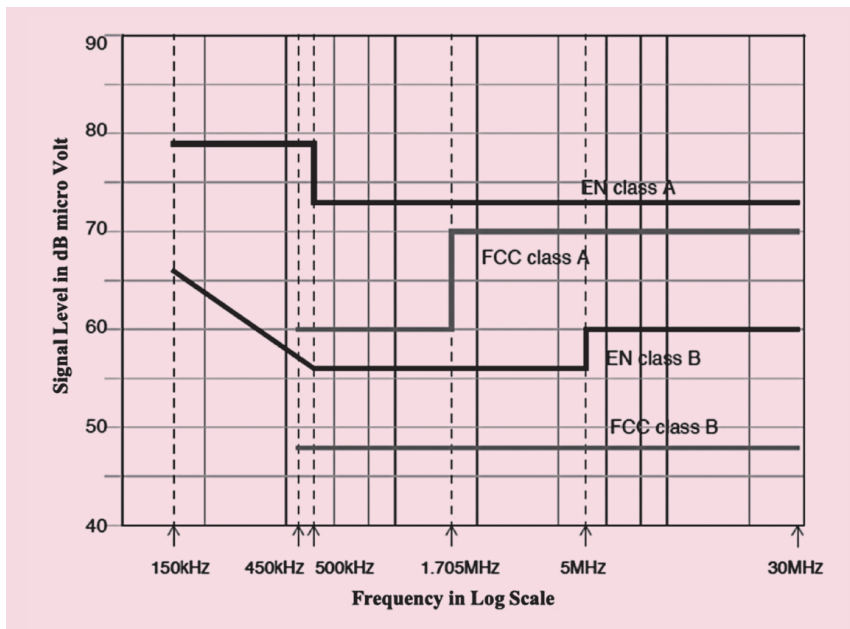


Fig. 3: Radiated and conducted emission levels of EU standards

Applicability of the Classes in USA and EU

Table 1

Compliance class	USA	European Union
Class A	Non-residential	Industrial
Class B	Residential	Residential and commercial

and a receiver for energy transfer. Typically, during testing, the transmitter would be the product under test and the receiver would be a spectrum analyser fitted with an antenna to measure strength of the RF energy. Fig. 1 shows the radiated and conduct-

ed emission mechanism in a product for our understanding.

But most radiated emissions are through apertures (such as ventilation holes, corner seams, and gaps between the chassis bottom and top) in the enclosure in addition to cables

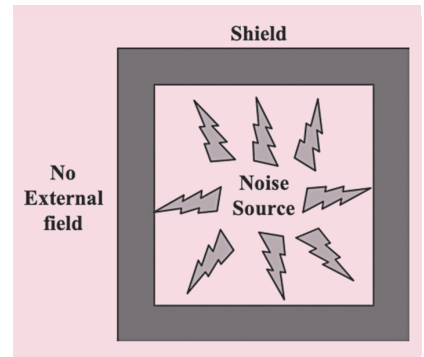


Fig. 4: Electronics system in a shielded enclosure with no external field

(power and signal) that add to the overall radiation. While power cable radiation is measured as CE radiation due to signal cables, it is a part of the total radiated emission. This is one reason why products with large cable connections (Ethernet switch cables, sensor cables, etc) have to be designed carefully and any source of radiation has to be contained within the enclosure, as managing cable radiation can be challenging and expensive.

For validation of a product's compliance, it is kept in an anechoic chamber and an antenna is kept at a designated distance from the product (three metres or ten metres) depending on the standards to which the product should comply. The antenna is connected to a spectrum analyser through proper instrumentation to measure level of signal radiating from the product for RE measurement. The reading is mapped against the level specified by the standards. If the signal level is lower than the specified level, the product is certified for compliance.

In case of conducted emission, power cable of product under test is connected through a line impedance stabilisation network (LISN) to spectrum analyser and the signal levels are measured and, as in case of radiated emission, mapped against the standard specified values. Fig. 2 and Fig. 3 show the radiated and conducted emission levels of FCC and EU standards mapped for understanding the standards better.

The standards define two sets of levels called Class A and Class B. Table 1 shows the applicability of the classes in the USA and EU.